ASSIGNMENT 7

# Sample Data Insertion

Project: Airline Reservation System | Course: Database Management Systems

**Objectives covered**

- Insert realistic sample data into all tables of the Airline Reservation System
- Include enough records to simulate a real-world application
- Ensure the data obeys the defined constraints and relationships

## 1. Introduction

This assignment populates the database created in Assignment 5 with realistic sample data using **INSERT** statements. The data models a small airline booking platform operating domestic Indian routes and a few international routes (Dubai, Singapore, London) in the weeks around **20 October – 9 November 2026**. It includes completed, scheduled, delayed and cancelled flights, confirmed, pending and cancelled bookings, and successful, failed and refunded payments, so that the queries in Assignments 8 and 9 produce meaningful results. All names, e-mail addresses, phone numbers and passport numbers are **fictional**.

**Note on realism:** airline names and airport codes are real, but the flight numbers, schedules, fares and passengers are invented for the project. Fares are in Indian Rupees (INR).

## 2. Insertion Order and Data Volume

Because of the foreign keys, parent tables must be filled before their children. The script runs with `PRAGMA foreign_keys = ON`, so any out-of-order or invalid row would have been rejected.

| Step | Table | Depends on | Rows | What the rows represent |
|---|---|---|---|---|
| 1 | airline | – | 6 | IndiGo, Air India, SpiceJet, Akasa Air, Emirates, Singapore Airlines |
| 2 | airport | – | 11 | 8 Indian airports and Dubai, Singapore, London |
| 3 | aircraft | airline | 11 | Fleet: A320, A321, 737, 787, 777, A350, Q400 (one in maintenance) |
| 4 | flight | aircraft, airport | 20 | 17 upcoming flights, 2 completed, 1 cancelled, 1 delayed |
| 5 | passenger | – | 25 | Families, business travellers, foreign nationals, children |
| 6 | booking | passenger | 20 | 17 confirmed, 1 pending, 2 cancelled |
| 7 | ticket | booking, flight, passenger | 30 | Economy, Premium Economy and Business seats |
| 8 | payment | booking | 23 | Successful, failed attempts and refunds |

Total: **146 rows** in 8 tables.

**Row counts after loading**

```
SELECT 1 AS step, 'airline' AS table_name, COUNT(*) AS rows_inserted FROM airline UNION ALL
SELECT 2, 'airport', COUNT(*) FROM airport UNION ALL SELECT 3, 'aircraft', COUNT(*) FROM aircraft UNION ALL
SELECT 4, 'flight', COUNT(*) FROM flight UNION ALL SELECT 5, 'passenger', COUNT(*) FROM passenger UNION ALL
SELECT 6, 'booking', COUNT(*) FROM booking UNION ALL SELECT 7, 'ticket', COUNT(*) FROM ticket UNION ALL
SELECT 8, 'payment', COUNT(*) FROM payment;
```

**OUTPUT**

```
+------+------------+---------------+
| step | table_name | rows_inserted |
+------+------------+---------------+
|    1 | airline    |             6 |
|    2 | airport    |            11 |
|    3 | aircraft   |            11 |
|    4 | flight     |            20 |
|    5 | passenger  |            25 |
|    6 | booking    |            20 |
|    7 | ticket     |            30 |
|    8 | payment    |            23 |
+------+------------+---------------+
8 rows in set
```

# 3. INSERT Statements

The complete insertion script is shown below, one section per table. (The same script is supplied as *data.sql*.)

## 3.1 airline

Six airlines with their 2-letter IATA codes.

**INSERT INTO airline**

```
INSERT INTO airline (airline_id, airline_name, iata_code, country) VALUES
    (1, 'IndiGo', '6E', 'India'),
    (2, 'Air India', 'AI', 'India'),
    (3, 'SpiceJet', 'SG', 'India'),
    (4, 'Akasa Air', 'QP', 'India'),
    (5, 'Emirates', 'EK', 'United Arab Emirates'),
    (6, 'Singapore Airlines', 'SQ', 'Singapore');
```

## 3.2 airport

Eleven airports. GOI (Goa) is deliberately not used by any flight yet, which is useful for outer-join queries.

**INSERT INTO airport**

```
INSERT INTO airport (airport_code, airport_name, city, country) VALUES
    ('BLR', 'Kempegowda International Airport', 'Bengaluru', 'India'),
    ('DEL', 'Indira Gandhi International Airport', 'New Delhi', 'India'),
    ('BOM', 'Chhatrapati Shivaji Maharaj International Airport', 'Mumbai', 'India'),
    ('MAA', 'Chennai International Airport', 'Chennai', 'India'),
    ('HYD', 'Rajiv Gandhi International Airport', 'Hyderabad', 'India'),
    ('CCU', 'Netaji Subhas Chandra Bose International Airport', 'Kolkata', 'India'),
    ('COK', 'Cochin International Airport', 'Kochi', 'India'),
    ('GOI', 'Dabolim Airport', 'Goa', 'India'),
    ('DXB', 'Dubai International Airport', 'Dubai', 'United Arab Emirates'),
    ('SIN', 'Singapore Changi Airport', 'Singapore', 'Singapore'),
    ('LHR', 'London Heathrow Airport', 'London', 'United Kingdom');
```

## 3.3 aircraft

Eleven aircraft with realistic seat capacities. VT-SPQ is in maintenance and has no flights, which is useful for outer-join queries.

```
INSERT INTO aircraft
INSERT INTO aircraft (
    aircraft_id, registration_no, model, total_seats, status, airline_id
) VALUES
    (1, 'VT-IJA', 'Airbus A320neo', 186, 'Active', 1),
    (2, 'VT-IMB', 'Airbus A321neo', 222, 'Active', 1),
    (3, 'VT-AJC', 'Boeing 787-8 Dreamliner', 238, 'Active', 2),
    (4, 'VT-TNA', 'Airbus A320', 168, 'Active', 2),
    (5, 'VT-SZB', 'Boeing 737-800', 189, 'Active', 3),
    (6, 'VT-YAC', 'Boeing 737-8 MAX', 189, 'Active', 4),
    (7, 'A6-ENX', 'Boeing 777-300ER', 354, 'Active', 5),
    (8, '9V-SMD', 'Airbus A350-900', 253, 'Active', 6),
    (9, 'VT-IJZ', 'Airbus A320neo', 186, 'Active', 1),
    (10, 'VT-AIQ', 'Airbus A321', 182, 'Active', 2),
    (11, 'VT-SPQ', 'Bombardier Q400', 78, 'Maintenance', 3);
```

## 3.4 flight

Twenty flights. Some use the same flight number on different dates (6E-501, AI-807, EK-565). Two are in the past (Completed), one is Delayed and one Cancelled.

```
INSERT INTO flight
INSERT INTO flight (
    flight_id, flight_number, aircraft_id, origin_code, destination_code, departure_time,
    arrival_time, base_fare, status
) VALUES
    (1, '6E-501', 1, 'BLR', 'DEL', '2026-11-02 06:00:00', '2026-11-02 08:45:00', 5200, 'Scheduled'),
    (2, '6E-502', 1, 'DEL', 'BLR', '2026-11-02 10:00:00', '2026-11-02 12:50:00', 5100, 'Scheduled'),
    (3, 'AI-807', 4, 'BLR', 'BOM', '2026-11-02 07:30:00', '2026-11-02 09:30:00', 4300, 'Scheduled'),
    (4, 'AI-808', 4, 'BOM', 'BLR', '2026-11-02 11:00:00', '2026-11-02 13:05:00', 4200, 'Delayed'),
    (5, 'SG-612', 5, 'DEL', 'BOM', '2026-11-03 09:15:00', '2026-11-03 11:25:00', 3900, 'Scheduled'),
    (6, 'QP-1305', 6, 'BLR', 'HYD', '2026-11-03 08:00:00', '2026-11-03 09:20:00', 2800, 'Scheduled'),
    (7, '6E-231', 9, 'BLR', 'MAA', '2026-11-03 18:30:00', '2026-11-03 19:35:00', 2500, 'Scheduled'),
    (8, 'AI-505', 10, 'DEL', 'CCU', '2026-11-04 14:00:00', '2026-11-04 16:20:00', 4600, 'Scheduled'),
    (9, 'EK-565', 7, 'BLR', 'DXB', '2026-11-04 03:50:00', '2026-11-04 06:20:00', 21500, 'Scheduled'),
    (10, 'SQ-503', 8, 'MAA', 'SIN', '2026-11-03 22:10:00', '2026-11-04 05:00:00', 24800,
      'Scheduled'),
    (11, 'AI-111', 3, 'DEL', 'LHR', '2026-11-05 02:00:00', '2026-11-05 05:45:00', 52000,
      'Scheduled'),
    (12, '6E-714', 2, 'BOM', 'BLR', '2026-11-06 20:00:00', '2026-11-06 22:10:00', 4000, 'Scheduled'),
    (13, 'SG-148', 5, 'BOM', 'HYD', '2026-11-06 12:45:00', '2026-11-06 14:15:00', 3300, 'Scheduled'),
    (14, 'QP-1410', 6, 'HYD', 'DEL', '2026-11-06 17:30:00', '2026-11-06 19:45:00', 4100,
      'Scheduled'),
    (15, 'EK-567', 7, 'DXB', 'BLR', '2026-11-08 09:30:00', '2026-11-08 14:00:00', 22300,
      'Scheduled'),
    (16, '6E-319', 2, 'BLR', 'COK', '2026-11-08 16:30:00', '2026-11-08 17:50:00', 3200, 'Scheduled'),
    (17, 'AI-509', 4, 'DEL', 'BLR', '2026-11-09 19:00:00', '2026-11-09 21:50:00', 5400, 'Scheduled'),
    (18, '6E-501', 1, 'BLR', 'DEL', '2026-10-20 06:00:00', '2026-10-20 08:45:00', 5000, 'Completed'),
    (19, 'AI-807', 4, 'BLR', 'BOM', '2026-10-21 07:30:00', '2026-10-21 09:30:00', 4100, 'Completed'),
    (20, 'EK-565', 7, 'BLR', 'DXB', '2026-10-30 03:50:00', '2026-10-30 06:20:00', 21000,
      'Cancelled');
```

## 3.5 passenger

Twenty-five passengers. Ages range from 10 to 50 years; two are foreign nationals. Passenger 25 has not booked anything yet.

INSERT INTO passenger

```
INSERT INTO passenger (
    passenger_id, first_name, last_name, email, phone, date_of_birth, gender, passport_no,
    nationality, registered_on
) VALUES
    (1, 'Aarav', 'Sharma', 'aarav.sharma@example.com', '+91-6399869879', '1990-03-14', 'M',
      'K6360060', 'Indian', '2025-03-27'),
    (2, 'Priya', 'Nair', 'priya.nair@example.com', '+91-6141707536', '1988-07-22', 'F', 'M1042527',
      'Indian', '2024-09-14'),
    (3, 'Rohan', 'Mehta', 'rohan.mehta@example.com', '+91-9815188447', '1995-11-05', 'M',
      'P6983360', 'Indian', '2024-05-24'),
    (4, 'Ananya', 'Iyer', 'ananya.iyer@example.com', '+91-6557616987', '1999-01-30', 'F',
      'Z6184138', 'Indian', '2026-06-07'),
    (5, 'Vikram', 'Singh', 'vikram.singh@example.com', '+91-9684697880', '1982-09-17', 'M',
      'K8137754', 'Indian', '2025-08-12'),
    (6, 'Sneha', 'Reddy', 'sneha.reddy@example.com', '+91-6819672731', '1985-04-09', 'F',
      'M3164241', 'Indian', '2025-10-03'),
    (7, 'Arjun', 'Reddy', 'arjun.reddy@example.com', '+91-8864887663', '2012-06-21', 'M',
      'S8875326', 'Indian', '2024-10-28'),
    (8, 'Kavya', 'Reddy', 'kavya.reddy@example.com', '+91-6297875347', '2015-12-02', 'F',
      'N1155640', 'Indian', '2025-04-22'),
    (9, 'Mohammed', 'Faizan', 'mohammed.faizan@example.com', '+91-9804571784', '1992-08-12', 'M',
      'T9340912', 'Indian', '2025-04-08'),
    (10, 'Divya', 'Krishnan', 'divya.krishnan@example.com', '+91-9147513627', '1993-02-27', 'F',
      'L8143344', 'Indian', '2024-01-15'),
    (11, 'Rahul', 'Verma', 'rahul.verma@example.com', '+91-9981933190', '1987-10-03', 'M',
      'K1003589', 'Indian', '2026-02-08'),
    (12, 'Neha', 'Verma', 'neha.verma@example.com', '+91-8205997283', '1989-05-19', 'F',
      'T5505293', 'Indian', '2025-04-29'),
    (13, 'Suresh', 'Kumar', 'suresh.kumar@example.com', '+91-8162294866', '1975-12-25', 'M',
      'U3201443', 'Indian', '2026-04-07'),
    (14, 'Lakshmi', 'Menon', 'lakshmi.menon@example.com', '+91-7601610550', '1979-03-08', 'F',
      'L9724671', 'Indian', '2024-01-27'),
    (15, 'Gautham', 'Menon', 'gautham.menon@example.com', '+91-8475177992', '2010-09-14', 'M',
      'T8986364', 'Indian', '2024-06-06'),
    (16, 'Pooja', 'Desai', 'pooja.desai@example.com', '+91-7695793186', '1994-06-01', 'F',
      'U9717671', 'Indian', '2024-02-04'),
    (17, 'Sanjay', 'Gupta', 'sanjay.gupta@example.com', '+91-6807370283', '1980-01-11', 'M',
      'M3908563', 'Indian', '2026-03-15'),
    (18, 'Meera', 'Gupta', 'meera.gupta@example.com', '+91-7264686164', '1983-07-30', 'F',
      'T3310587', 'Indian', '2024-07-10'),
    (19, 'David', 'Thomas', 'david.thomas@example.com', '+91-8776472587', '1991-04-23', 'M',
      'P3930894', 'Indian', '2025-03-31'),
    (20, 'Emily', 'Thomas', 'emily.thomas@example.com', '+91-8133017568', '1992-11-16', 'F',
      'K9845818', 'British', '2026-03-13'),
    (21, 'Karthik', 'Subramanian', 'karthik.subramanian@example.com', '+91-6525898319',
      '1978-02-05', 'M', 'Z4061027', 'Indian', '2026-01-03'),
    (22, 'Farah', 'Khan', 'farah.khan@example.com', '+91-8948214162', '1996-09-09', 'F',
      'N4597623', 'Indian', '2024-04-18'),
    (23, 'Amit', 'Patel', 'amit.patel@example.com', '+91-8218549145', '1986-08-28', 'M',
      'Z2013781', 'Indian', '2024-06-14'),
    (24, 'Ishita', 'Bose', 'ishita.bose@example.com', '+91-6064600130', '2001-03-03', 'F',
      'N5297049', 'Indian', '2024-11-13'),
    (25, 'Tenzin', 'Dorjee', 'tenzin.dorjee@example.com', '+91-8579906593', '1997-12-12', 'M',
      'K2574879', 'Bhutanese', '2025-08-31');
```

## 3.6 booking

Twenty bookings by passengers. total_amount is not supplied; it takes its DEFAULT value 0 and is computed from the tickets in step 3.7.

INSERT INTO booking

```
INSERT INTO booking (booking_id, booking_ref, passenger_id, booking_date, status) VALUES
    (1, '75HZS2', 1, '2026-10-01 16:52:00', 'Confirmed'),
    (2, 'D5BUXU', 2, '2026-10-03 10:19:00', 'Confirmed'),
    (3, 'BXNDHL', 3, '2026-10-10 10:07:00', 'Confirmed'),
    (4, 'YVB5JY', 5, '2026-10-11 20:24:00', 'Confirmed'),
    (5, 'R4QJMB', 6, '2026-10-12 19:56:00', 'Confirmed'),
    (6, 'BBT84S', 9, '2026-10-13 17:08:00', 'Confirmed'),
    (7, 'FMCSXW', 10, '2026-10-14 08:42:00', 'Confirmed'),
    (8, 'R8DQK3', 11, '2026-10-15 18:50:00', 'Confirmed'),
    (9, 'TXNZY7', 13, '2026-10-16 21:35:00', 'Cancelled'),
    (10, 'VNL4K9', 14, '2026-10-17 07:22:00', 'Confirmed'),
    (11, 'CMFVZS', 16, '2026-10-18 11:45:00', 'Confirmed'),
    (12, 'BPVC32', 17, '2026-10-19 12:32:00', 'Confirmed'),
    (13, 'K2XSVC', 19, '2026-10-19 13:10:00', 'Confirmed'),
    (14, 'UYJ2NA', 21, '2026-10-20 22:27:00', 'Confirmed'),
    (15, 'LBMWKL', 22, '2026-10-21 07:56:00', 'Pending'),
    (16, 'AZEZVP', 1, '2026-10-21 11:38:00', 'Confirmed'),
    (17, 'JGW9GE', 2, '2026-10-22 19:16:00', 'Confirmed'),
    (18, 'HY89HR', 4, '2026-10-22 18:43:00', 'Confirmed'),
    (19, 'YFEF9V', 24, '2026-10-23 15:48:00', 'Confirmed'),
    (20, 'LW5XW9', 8, '2026-10-23 19:20:00', 'Cancelled');
```

## 3.7 ticket and booking totals

Thirty tickets. A booking may hold several tickets (family trips, colleagues), and a passenger may travel on a booking made by someone else. Cancelled tickets have seat_no = NULL so that the seat is free again. After inserting, an UPDATE with a correlated subquery fills booking.total_amount.

INSERT INTO ticket + UPDATE

```
INSERT INTO ticket (
    ticket_id, booking_id, flight_id, passenger_id, seat_no, travel_class, fare, status
) VALUES
    (1, 1, 18, 1, '12A', 'Economy', 5000, 'Flown'),
    (2, 2, 19, 2, '14C', 'Economy', 4510, 'Flown'),
    (3, 2, 19, 3, '14D', 'Economy', 4510, 'Flown'),
    (4, 3, 1, 3, '8A', 'Economy', 4680, 'Booked'),
    (5, 3, 1, 4, '8B', 'Economy', 4680, 'Booked'),
    (6, 4, 1, 5, '2A', 'Business', 16020, 'Booked'),
    (7, 5, 2, 6, '15F', 'Economy', 5100, 'Booked'),
    (8, 5, 2, 7, '15E', 'Economy', 5100, 'Booked'),
    (9, 5, 2, 8, '15D', 'Economy', 5100, 'Booked'),
    (10, 6, 3, 9, '10A', 'Economy', 5160, 'Booked'),
    (11, 7, 4, 10, '6C', 'Premium Economy', 5670, 'Booked'),
    (12, 8, 5, 11, '20A', 'Economy', 3900, 'Booked'),
    (13, 8, 5, 12, '20B', 'Economy', 3900, 'Booked'),
    (14, 9, 6, 13, NULL, 'Economy', 3080, 'Cancelled'),
    (15, 10, 7, 14, '11D', 'Economy', 3000, 'Booked'),
    (16, 10, 7, 15, '11E', 'Economy', 3000, 'Booked'),
    (17, 11, 8, 16, '5A', 'Premium Economy', 6210, 'Booked'),
    (18, 12, 9, 17, '3A', 'Business', 54180, 'Booked'),
    (19, 12, 9, 18, '3B', 'Business', 54180, 'Booked'),
    (20, 13, 10, 19, '30A', 'Economy', 24800, 'Booked'),
    (21, 13, 10, 20, '30B', 'Economy', 24800, 'Booked'),
    (22, 14, 11, 21, '4A', 'Business', 145600, 'Booked'),
    (23, 15, 12, 22, '18C', 'Economy', 3600, 'Booked'),
    (24, 16, 13, 1, '7B', 'Economy', 3960, 'Booked'),
    (25, 16, 14, 1, '12A', 'Economy', 4100, 'Booked'),
    (26, 17, 15, 2, '25C', 'Economy', 26760, 'Booked'),
    (27, 17, 15, 3, '25D', 'Economy', 26760, 'Booked'),
    (28, 18, 16, 4, '6A', 'Economy', 3520, 'Booked'),
    (29, 19, 17, 24, '9A', 'Premium Economy', 8100, 'Booked'),
    (30, 20, 20, 8, NULL, 'Economy', 21000, 'Cancelled');

-- booking totals derived from tickets
UPDATE booking
SET total_amount = (SELECT SUM(fare) FROM ticket WHERE ticket.booking_id = booking.booking_id);
```

### 3.8 payment

Twenty-three payment records: one successful payment per paid booking, two failed attempts for the pending booking, and a separate 'Refunded' row for each cancelled booking.

```
INSERT INTO payment
INSERT INTO payment (
    payment_id, booking_id, amount, method, payment_date, status, transaction_ref
) VALUES
    (1, 1, 5000, 'Debit Card', '2026-10-01 17:01:00', 'Success', 'TXN355696327'),
    (2, 2, 9020, 'Wallet', '2026-10-03 10:23:00', 'Success', 'TXN645154742'),
    (3, 3, 9360, 'Wallet', '2026-10-10 10:13:00', 'Success', 'TXN352814497'),
    (4, 4, 16020, 'Net Banking', '2026-10-11 20:32:00', 'Success', 'TXN085651788'),
    (5, 5, 15300, 'Debit Card', '2026-10-12 20:04:00', 'Success', 'TXN550620484'),
    (6, 6, 5160, 'Debit Card', '2026-10-13 17:15:00', 'Success', 'TXN996775003'),
    (7, 7, 5670, 'Credit Card', '2026-10-14 08:48:00', 'Success', 'TXN503784951'),
    (8, 8, 7800, 'Wallet', '2026-10-15 18:55:00', 'Success', 'TXN801253739'),
    (9, 9, 3080, 'Debit Card', '2026-10-16 21:41:00', 'Success', 'TXN196898210'),
    (10, 9, 3080, 'Debit Card', '2026-10-18 21:35:00', 'Refunded', 'TXN602883755'),
    (11, 10, 6000, 'UPI', '2026-10-17 07:29:00', 'Success', 'TXN974754072'),
    (12, 11, 6210, 'Credit Card', '2026-10-18 11:54:00', 'Success', 'TXN547354864'),
    (13, 12, 108360, 'Debit Card', '2026-10-19 12:34:00', 'Success', 'TXN618847918'),
    (14, 13, 49600, 'Wallet', '2026-10-19 13:17:00', 'Success', 'TXN755521824'),
    (15, 14, 145600, 'Wallet', '2026-10-20 22:35:00', 'Success', 'TXN193073368'),
    (16, 15, 3600, 'Net Banking', '2026-10-21 07:59:00', 'Failed', 'TXN828347285'),
    (17, 15, 3600, 'UPI', '2026-10-21 08:03:00', 'Failed', 'TXN568983298'),
    (18, 16, 8060, 'Debit Card', '2026-10-21 11:42:00', 'Success', 'TXN290044224'),
    (19, 17, 53520, 'Wallet', '2026-10-22 19:23:00', 'Success', 'TXN272202269'),
    (20, 18, 3520, 'Net Banking', '2026-10-22 18:45:00', 'Success', 'TXN438581762'),
    (21, 19, 8100, 'UPI', '2026-10-23 15:52:00', 'Success', 'TXN807001594'),
    (22, 20, 21000, 'UPI', '2026-10-23 19:22:00', 'Success', 'TXN283166323'),
    (23, 20, 21000, 'UPI', '2026-10-24 19:20:00', 'Refunded', 'TXN735286675');
```

# 4. Verification of the Inserted Data

The script finished without any error, which already proves that every row satisfied the PRIMARY KEY, UNIQUE, CHECK, NOT NULL and FOREIGN KEY constraints. The contents of the tables are displayed below with SELECT statements.

### 4.1 airline

```
SQL
SELECT * FROM airline;
```

```
OUTPUT
+------------+--------------------+-----------+----------------------+
| airline_id | airline_name       | iata_code | country              |
+------------+--------------------+-----------+----------------------+
|          1 | IndiGo             | 6E        | India                |
|          2 | Air India          | AI        | India                |
|          3 | SpiceJet           | SG        | India                |
|          4 | Akasa Air          | QP        | India                |
|          5 | Emirates           | EK        | United Arab Emirates |
|          6 | Singapore Airlines | SQ        | Singapore            |
+------------+--------------------+-----------+----------------------+
6 rows in set
```

### 4.2 airport

```sql
SELECT * FROM airport;
```

**OUTPUT**

| airport_code | airport_name | city | country |
|---|---|---|---|
| BLR | Kempegowda International Airport | Bengaluru | India |
| DEL | Indira Gandhi International Airport | New Delhi | India |
| BOM | Chhatrapati Shivaji Maharaj International Airport | Mumbai | India |
| MAA | Chennai International Airport | Chennai | India |
| HYD | Rajiv Gandhi International Airport | Hyderabad | India |
| CCU | Netaji Subhas Chandra Bose International Airport | Kolkata | India |
| COK | Cochin International Airport | Kochi | India |
| GOI | Dabolim Airport | Goa | India |
| DXB | Dubai International Airport | Dubai | United Arab Emirates |
| SIN | Singapore Changi Airport | Singapore | Singapore |
| LHR | London Heathrow Airport | London | United Kingdom |

11 rows in set

### 4.3 aircraft

```sql
SELECT * FROM aircraft;
```

**OUTPUT**

| aircraft_id | registration_no | model | total_seats | status | airline_id |
|---|---|---|---|---|---|
| 1 | VT-IJA | Airbus A320neo | 186 | Active | 1 |
| 2 | VT-IMB | Airbus A321neo | 222 | Active | 1 |
| 3 | VT-AJC | Boeing 787-8 Dreamliner | 238 | Active | 2 |
| 4 | VT-TNA | Airbus A320 | 168 | Active | 2 |
| 5 | VT-SZB | Boeing 737-800 | 189 | Active | 3 |
| 6 | VT-YAC | Boeing 737-8 MAX | 189 | Active | 4 |
| 7 | A6-ENX | Boeing 777-300ER | 354 | Active | 5 |
| 8 | 9V-SMD | Airbus A350-900 | 253 | Active | 6 |
| 9 | VT-IJZ | Airbus A320neo | 186 | Active | 1 |
| 10 | VT-AIQ | Airbus A321 | 182 | Active | 2 |
| 11 | VT-SPQ | Bombardier Q400 | 78 | Maintenance | 3 |

11 rows in set

## 4.4 flight

```
SQL
SELECT flight_id, flight_number, aircraft_id AS ac, origin_code AS org, destination_code AS dst, departure_time, arrival_time, base_fare, status FROM flight;
```

OUTPUT

| flight_id | flight_number | ac | org | dst | departure_time | arrival_time | base_fare | status |
|---|---|---|---|---|---|---|---|---|
| 1 | 6E-501 | 1 | BLR | DEL | 2026-11-02 06:00:00 | 2026-11-02 08:45:00 | 5200 | Scheduled |
| 2 | 6E-502 | 1 | DEL | BLR | 2026-11-02 10:00:00 | 2026-11-02 12:50:00 | 5100 | Scheduled |
| 3 | AI-807 | 4 | BLR | BOM | 2026-11-02 07:30:00 | 2026-11-02 09:30:00 | 4300 | Scheduled |
| 4 | AI-808 | 4 | BOM | BLR | 2026-11-02 11:00:00 | 2026-11-02 13:05:00 | 4200 | Delayed |
| 5 | SG-612 | 5 | DEL | BOM | 2026-11-03 09:15:00 | 2026-11-03 11:25:00 | 3900 | Scheduled |
| 6 | QP-1305 | 6 | BLR | HYD | 2026-11-03 08:00:00 | 2026-11-03 09:20:00 | 2800 | Scheduled |
| 7 | 6E-231 | 9 | BLR | MAA | 2026-11-03 18:30:00 | 2026-11-03 19:35:00 | 2500 | Scheduled |
| 8 | AI-505 | 10 | DEL | CCU | 2026-11-04 14:00:00 | 2026-11-04 16:20:00 | 4600 | Scheduled |
| 9 | EK-565 | 7 | BLR | DXB | 2026-11-04 03:50:00 | 2026-11-04 06:20:00 | 21500 | Scheduled |
| 10 | SQ-503 | 8 | MAA | SIN | 2026-11-03 22:10:00 | 2026-11-04 05:00:00 | 24800 | Scheduled |
| 11 | AI-111 | 3 | DEL | LHR | 2026-11-05 02:00:00 | 2026-11-05 05:45:00 | 52000 | Scheduled |
| 12 | 6E-714 | 2 | BOM | BLR | 2026-11-06 20:00:00 | 2026-11-06 22:10:00 | 4000 | Scheduled |
| 13 | SG-148 | 5 | BOM | HYD | 2026-11-06 12:45:00 | 2026-11-06 14:15:00 | 3300 | Scheduled |
| 14 | QP-1410 | 6 | HYD | DEL | 2026-11-06 17:30:00 | 2026-11-06 19:45:00 | 4100 | Scheduled |
| 15 | EK-567 | 7 | DXB | BLR | 2026-11-08 09:30:00 | 2026-11-08 14:00:00 | 22300 | Scheduled |
| 16 | 6E-319 | 2 | BLR | COK | 2026-11-08 16:30:00 | 2026-11-08 17:50:00 | 3200 | Scheduled |
| 17 | AI-509 | 4 | DEL | BLR | 2026-11-09 19:00:00 | 2026-11-09 21:50:00 | 5400 | Scheduled |
| 18 | 6E-501 | 1 | BLR | DEL | 2026-10-20 06:00:00 | 2026-10-20 08:45:00 | 5000 | Completed |
| 19 | AI-807 | 4 | BLR | BOM | 2026-10-21 07:30:00 | 2026-10-21 09:30:00 | 4100 | Completed |
| 20 | EK-565 | 7 | BLR | DXB | 2026-10-30 03:50:00 | 2026-10-30 06:20:00 | 21000 | Cancelled |

20 rows in set

## 4.5 passenger

```
SQL
SELECT passenger_id AS id, first_name, last_name, email, phone, date_of_birth AS dob, gender AS g, nationality FROM passenger;
```

OUTPUT

| id | first_name | last_name | email | phone | dob | g | nationality |
|---|---|---|---|---|---|---|---|
| 1 | Aarav | Sharma | aarav.sharma@example.com | +91-6399869879 | 1990-03-14 | M | Indian |
| 2 | Priya | Nair | priya.nair@example.com | +91-6141707536 | 1988-07-22 | F | Indian |
| 3 | Rohan | Mehta | rohan.mehta@example.com | +91-9815188447 | 1995-11-05 | M | Indian |
| 4 | Ananya | Iyer | ananya.iyer@example.com | +91-6557616987 | 1999-01-30 | F | Indian |
| 5 | Vikram | Singh | vikram.singh@example.com | +91-9684697880 | 1982-09-17 | M | Indian |
| 6 | Sneha | Reddy | sneha.reddy@example.com | +91-6819672731 | 1985-04-09 | F | Indian |
| 7 | Arjun | Reddy | arjun.reddy@example.com | +91-8864887663 | 2012-06-21 | M | Indian |
| 8 | Kavya | Reddy | kavya.reddy@example.com | +91-6297875347 | 2015-12-02 | F | Indian |
| 9 | Mohammed | Faizan | mohammed.faizan@example.com | +91-9804571784 | 1992-08-12 | M | Indian |
| 10 | Divya | Krishnan | divya.krishnan@example.com | +91-9147513627 | 1993-02-27 | F | Indian |
| 11 | Rahul | Verma | rahul.verma@example.com | +91-9981933190 | 1987-10-03 | M | Indian |
| 12 | Neha | Verma | neha.verma@example.com | +91-8205997283 | 1989-05-19 | F | Indian |
| 13 | Suresh | Kumar | suresh.kumar@example.com | +91-8162294866 | 1975-12-25 | M | Indian |
| 14 | Lakshmi | Menon | lakshmi.menon@example.com | +91-7601610550 | 1979-03-08 | F | Indian |
| 15 | Gautham | Menon | gautham.menon@example.com | +91-8475177992 | 2010-09-14 | M | Indian |
| 16 | Pooja | Desai | pooja.desai@example.com | +91-7695793186 | 1994-06-01 | F | Indian |
| 17 | Sanjay | Gupta | sanjay.gupta@example.com | +91-6807370283 | 1980-01-11 | M | Indian |
| 18 | Meera | Gupta | meera.gupta@example.com | +91-7264686164 | 1983-07-30 | F | Indian |
| 19 | David | Thomas | david.thomas@example.com | +91-8776472587 | 1991-04-23 | M | Indian |
| 20 | Emily | Thomas | emily.thomas@example.com | +91-8133017568 | 1992-11-16 | F | British |
| 21 | Karthik | Subramanian | karthik.subramanian@example.com | +91-6525898319 | 1978-02-05 | M | Indian |
| 22 | Farah | Khan | farah.khan@example.com | +91-8948214162 | 1996-09-09 | F | Indian |
| 23 | Amit | Patel | amit.patel@example.com | +91-8218549145 | 1986-08-28 | M | Indian |
| 24 | Ishita | Bose | ishita.bose@example.com | +91-6064600130 | 2001-03-03 | F | Indian |
| 25 | Tenzin | Dorjee | tenzin.dorjee@example.com | +91-8579906593 | 1997-12-12 | M | Bhutanese |

25 rows in set

### 4.6 booking

```sql
SELECT * FROM booking;
```

**OUTPUT**

| booking_id | booking_ref | passenger_id | booking_date | status | total_amount |
|---|---|---|---|---|---|
| 1 | 75HZS2 | 1 | 2026-10-01 16:52:00 | Confirmed | 5000 |
| 2 | D5BUXU | 2 | 2026-10-03 10:19:00 | Confirmed | 9020 |
| 3 | BXNDHL | 3 | 2026-10-10 10:07:00 | Confirmed | 9360 |
| 4 | YVB5JY | 5 | 2026-10-11 20:24:00 | Confirmed | 16020 |
| 5 | R4QJMB | 6 | 2026-10-12 19:56:00 | Confirmed | 15300 |
| 6 | BBT84S | 9 | 2026-10-13 17:08:00 | Confirmed | 5160 |
| 7 | FMCSXW | 10 | 2026-10-14 08:42:00 | Confirmed | 5670 |
| 8 | R8DQK3 | 11 | 2026-10-15 18:50:00 | Confirmed | 7800 |
| 9 | TXNZY7 | 13 | 2026-10-16 21:35:00 | Cancelled | 3080 |
| 10 | VNL4K9 | 14 | 2026-10-17 07:22:00 | Confirmed | 6000 |
| 11 | CMFVZS | 16 | 2026-10-18 11:45:00 | Confirmed | 6210 |
| 12 | BPVC32 | 17 | 2026-10-19 12:32:00 | Confirmed | 108360 |
| 13 | K2XSVC | 19 | 2026-10-19 13:10:00 | Confirmed | 49600 |
| 14 | UYJ2NA | 21 | 2026-10-20 22:27:00 | Confirmed | 145600 |
| 15 | LBMWKL | 22 | 2026-10-21 07:56:00 | Pending | 3600 |
| 16 | AZEZVP | 1 | 2026-10-21 11:38:00 | Confirmed | 8060 |
| 17 | JGW9GE | 2 | 2026-10-22 19:16:00 | Confirmed | 53520 |
| 18 | HY89HR | 4 | 2026-10-22 18:43:00 | Confirmed | 3520 |
| 19 | YFEF9V | 24 | 2026-10-23 15:48:00 | Confirmed | 8100 |
| 20 | LW5XW9 | 8 | 2026-10-23 19:20:00 | Cancelled | 21000 |

20 rows in set

### 4.7 ticket

```sql
SELECT * FROM ticket;
```

**OUTPUT**

| ticket_id | booking_id | flight_id | passenger_id | seat_no | travel_class | fare | status |
|---|---|---|---|---|---|---|---|
| 1 | 1 | 18 | 1 | 12A | Economy | 5000 | Flown |
| 2 | 2 | 19 | 2 | 14C | Economy | 4510 | Flown |
| 3 | 2 | 19 | 3 | 14D | Economy | 4510 | Flown |
| 4 | 3 | 1 | 3 | 8A | Economy | 4680 | Booked |
| 5 | 3 | 1 | 4 | 8B | Economy | 4680 | Booked |
| 6 | 4 | 1 | 5 | 2A | Business | 16020 | Booked |
| 7 | 5 | 2 | 6 | 15F | Economy | 5100 | Booked |
| 8 | 5 | 2 | 7 | 15E | Economy | 5100 | Booked |
| 9 | 5 | 2 | 8 | 15D | Economy | 5100 | Booked |
| 10 | 6 | 3 | 9 | 10A | Economy | 5160 | Booked |
| 11 | 7 | 4 | 10 | 6C | Premium Economy | 5670 | Booked |
| 12 | 8 | 5 | 11 | 20A | Economy | 3900 | Booked |
| 13 | 8 | 5 | 12 | 20B | Economy | 3900 | Booked |
| 14 | 9 | 6 | 13 | NULL | Economy | 3080 | Cancelled |
| 15 | 10 | 7 | 14 | 11D | Economy | 3000 | Booked |
| 16 | 10 | 7 | 15 | 11E | Economy | 3000 | Booked |
| 17 | 11 | 8 | 16 | 5A | Premium Economy | 6210 | Booked |
| 18 | 12 | 9 | 17 | 3A | Business | 54180 | Booked |
| 19 | 12 | 9 | 18 | 3B | Business | 54180 | Booked |
| 20 | 13 | 10 | 19 | 30A | Economy | 24800 | Booked |
| 21 | 13 | 10 | 20 | 30B | Economy | 24800 | Booked |
| 22 | 14 | 11 | 21 | 4A | Business | 145600 | Booked |
| 23 | 15 | 12 | 22 | 18C | Economy | 3600 | Booked |
| 24 | 16 | 13 | 1 | 7B | Economy | 3960 | Booked |
| 25 | 16 | 14 | 1 | 12A | Economy | 4100 | Booked |
| 26 | 17 | 15 | 2 | 25C | Economy | 26760 | Booked |
| 27 | 17 | 15 | 3 | 25D | Economy | 26760 | Booked |
| 28 | 18 | 16 | 4 | 6A | Economy | 3520 | Booked |
| 29 | 19 | 17 | 24 | 9A | Premium Economy | 8100 | Booked |

```
|        30 |         20 |        20 |            8 | NULL    | Economy         |  21000 | Cancelled |
+-----------+------------+-----------+--------------+---------+-----------------+--------+-----------+
30 rows in set
```

### 4.8 payment

```sql
SELECT * FROM payment;
```

OUTPUT

```
+------------+------------+--------+-------------+---------------------+----------+-----------------+
| payment_id | booking_id | amount | method      | payment_date        | status   | transaction_ref |
+------------+------------+--------+-------------+---------------------+----------+-----------------+
|          1 |          1 |   5000 | Debit Card  | 2026-10-01 17:01:00 | Success  | TXN355696327    |
|          2 |          2 |   9020 | Wallet      | 2026-10-03 10:23:00 | Success  | TXN645154742    |
|          3 |          3 |   9360 | Wallet      | 2026-10-10 10:13:00 | Success  | TXN352814497    |
|          4 |          4 |  16020 | Net Banking | 2026-10-11 20:32:00 | Success  | TXN085651788    |
|          5 |          5 |  15300 | Debit Card  | 2026-10-12 20:04:00 | Success  | TXN550620484    |
|          6 |          6 |   5160 | Debit Card  | 2026-10-13 17:15:00 | Success  | TXN996775003    |
|          7 |          7 |   5670 | Credit Card | 2026-10-14 08:48:00 | Success  | TXN503784951    |
|          8 |          8 |   7800 | Wallet      | 2026-10-15 18:55:00 | Success  | TXN801253739    |
|          9 |          9 |   3080 | Debit Card  | 2026-10-16 21:41:00 | Success  | TXN196898210    |
|         10 |          9 |   3080 | Debit Card  | 2026-10-18 21:35:00 | Refunded | TXN602883755    |
|         11 |         10 |   6000 | UPI         | 2026-10-17 07:29:00 | Success  | TXN974754072    |
|         12 |         11 |   6210 | Credit Card | 2026-10-18 11:54:00 | Success  | TXN547354864    |
|         13 |         12 | 108360 | Debit Card  | 2026-10-19 12:34:00 | Success  | TXN618847918    |
|         14 |         13 |  49600 | Wallet      | 2026-10-19 13:17:00 | Success  | TXN755521824    |
|         15 |         14 | 145600 | Wallet      | 2026-10-20 22:35:00 | Success  | TXN193073368    |
|         16 |         15 |   3600 | Net Banking | 2026-10-21 07:59:00 | Failed   | TXN828347285    |
|         17 |         15 |   3600 | UPI         | 2026-10-21 08:03:00 | Failed   | TXN568983298    |
|         18 |         16 |   8060 | Debit Card  | 2026-10-21 11:42:00 | Success  | TXN290044224    |
|         19 |         17 |  53520 | Wallet      | 2026-10-22 19:23:00 | Success  | TXN272202269    |
|         20 |         18 |   3520 | Net Banking | 2026-10-22 18:45:00 | Success  | TXN438581762    |
|         21 |         19 |   8100 | UPI         | 2026-10-23 15:52:00 | Success  | TXN807001594    |
|         22 |         20 |  21000 | UPI         | 2026-10-23 19:22:00 | Success  | TXN283166323    |
|         23 |         20 |  21000 | UPI         | 2026-10-24 19:20:00 | Refunded | TXN735286675    |
+------------+------------+--------+-------------+---------------------+----------+-----------------+
23 rows in set
```

## 4.9 Totals and constraint compliance

**Booking totals versus ticket fares and successful payments**

```
SELECT booking_id, status, total_amount,
       (SELECT SUM(fare) FROM ticket t WHERE t.booking_id = b.booking_id) AS sum_of_fares,
       (SELECT COALESCE(SUM(amount),0) FROM payment p WHERE p.booking_id = b.booking_id AND p.status = 'Success') AS paid
FROM booking b ORDER BY booking_id;
```

**OUTPUT**

```
+------------+-----------+--------------+--------------+--------+
| booking_id | status    | total_amount | sum_of_fares | paid   |
+------------+-----------+--------------+--------------+--------+
|          1 | Confirmed |         5000 |         5000 |   5000 |
|          2 | Confirmed |         9020 |         9020 |   9020 |
|          3 | Confirmed |         9360 |         9360 |   9360 |
|          4 | Confirmed |        16020 |        16020 |  16020 |
|          5 | Confirmed |        15300 |        15300 |  15300 |
|          6 | Confirmed |         5160 |         5160 |   5160 |
|          7 | Confirmed |         5670 |         5670 |   5670 |
|          8 | Confirmed |         7800 |         7800 |   7800 |
|          9 | Cancelled |         3080 |         3080 |   3080 |
|         10 | Confirmed |         6000 |         6000 |   6000 |
|         11 | Confirmed |         6210 |         6210 |   6210 |
|         12 | Confirmed |       108360 |       108360 | 108360 |
|         13 | Confirmed |        49600 |        49600 |  49600 |
|         14 | Confirmed |       145600 |       145600 | 145600 |
|         15 | Pending   |         3600 |         3600 |      0 |
|         16 | Confirmed |         8060 |         8060 |   8060 |
|         17 | Confirmed |        53520 |        53520 |  53520 |
|         18 | Confirmed |         3520 |         3520 |   3520 |
|         19 | Confirmed |         8100 |         8100 |   8100 |
|         20 | Cancelled |        21000 |        21000 |  21000 |
+------------+-----------+--------------+--------------+--------+
20 rows in set
```

The computed totals equal the sum of the ticket fares for every booking, and every Confirmed or Cancelled booking has a successful payment equal to its total. The pending booking (15) has no successful payment yet.

**Orphan-record check**

```
SELECT (SELECT COUNT(*) FROM ticket  WHERE flight_id    NOT IN (SELECT flight_id    FROM flight))    AS tickets_without_flight,
       (SELECT COUNT(*) FROM ticket  WHERE booking_id   NOT IN (SELECT booking_id   FROM booking))   AS tickets_without_booking,
       (SELECT COUNT(*) FROM payment WHERE booking_id   NOT IN (SELECT booking_id   FROM booking))   AS payments_without_booking,
       (SELECT COUNT(*) FROM flight  WHERE aircraft_id  NOT IN (SELECT aircraft_id  FROM aircraft))  AS flights_without_aircraft;
PRAGMA foreign_key_check;
```

**OUTPUT**

```
+-----------------------+------------------------+--------------------------+--------------------------+
| tickets_without_flight | tickets_without_booking | payments_without_booking | flights_without_aircraft |
+-----------------------+------------------------+--------------------------+--------------------------+
|                     0 |                      0 |                        0 |                        0 |
+-----------------------+------------------------+--------------------------+--------------------------+
1 row in set
foreign_key_check -> 0 violation(s)
```

# 5. Data Characteristics (Realism Summary)

**SQL**

```
SELECT status, COUNT(*) AS flights FROM flight GROUP BY status ORDER BY flights DESC;
```

**OUTPUT**

```
+-----------+---------+
| status    | flights |
+-----------+---------+
| Scheduled |      16 |
| Completed |       2 |
| Delayed   |       1 |
| Cancelled |       1 |
+-----------+---------+
4 rows in set
```

```sql
SELECT travel_class, COUNT(*) AS tickets, MIN(fare) AS min_fare, MAX(fare) AS max_fare FROM ticket GROUP BY travel_class ORDER BY min_fare;
```

```
OUTPUT
+-----------------+---------+----------+----------+
| travel_class    | tickets | min_fare | max_fare |
+-----------------+---------+----------+----------+
| Economy         |      23 |     3000 |    26760 |
| Premium Economy |       3 |     5670 |     8100 |
| Business        |       4 |    16020 |   145600 |
+-----------------+---------+----------+----------+
3 rows in set
```

```sql
SELECT method, COUNT(*) AS payments FROM payment GROUP BY method ORDER BY payments DESC, method;
```

```
OUTPUT
+-------------+----------+
| method      | payments |
+-------------+----------+
| Debit Card  |        7 |
| Wallet      |        6 |
| UPI         |        5 |
| Net Banking |        3 |
| Credit Card |        2 |
+-------------+----------+
5 rows in set
```

- **Variety of states:** flights are Scheduled, Delayed, Completed or Cancelled; bookings are Confirmed, Pending or Cancelled; payments are Success, Failed or Refunded.
- **Edge cases for later queries:** an airport with no flights (GOI), an aircraft with no flights (VT-SPQ), a passenger who has never booked (Tenzin Dorjee), and a flight with a single ticket (6E-319).
- **Realistic business patterns:** family bookings with several travellers, a business-class booking on a long-haul route, connecting itineraries (BLR–MAA–SIN and BOM–HYD–DEL), repeat flight numbers on different dates, and fare differences by class.
- **Relationships verified:** every ticket references an existing booking, flight and passenger; every flight references an existing aircraft and two airports.

# 6. Conclusion

The database now holds 146 interrelated rows across all eight tables. The load script ran with foreign keys enforced and completed without a single constraint violation, and the verification queries show consistent totals and no orphan records. This dataset is the basis for the SQL queries and reports in Assignments 8 and 9.